Illumination Balance and Sharp Thresholds for Near-Field Depth Calibration

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Abstract

Near-field intensity can be identical at different depths because attenuation and incidence angle vary together. We derive the minimum coaxial-to-off-axis coefficient ratios that make the response strictly decrease at every positive depth, for a camera-axis point with fixed calibrated normal and gain. For isotropic illumination, the threshold is sharp over camera-plane source layouts and a prescribed normal cone. Weighted balance at a common radius removes the tilt penalty after sourcewise Lambertian clamping; a weighted-centroid residual quantifies imperfect balance. At the threshold, uniqueness can coexist with a non-Lipschitz inverse. A positive coefficient margin bounds finite-band depth error and yields fixed-budget allocations with calibration tolerances. For aligned angular emission, the threshold is determined by branch endpoints and roots of polynomials of degree at most four. It grows linearly with the normal-cone slope for each fixed positive emission exponent. Two allocations of a balanced common-radius array recover depth without normal or gain calibration when all incidence factors are nonnegative. With partial clamping, distinct depths can produce identical intensities for every mixture of the coaxial source and one fixed off-axis pattern. Separate forward and inverse calculations verify the formulas.

Keywords: Near-field illumination, depth calibration, photometric stereo, LED lighting, monotonicity, inverse problems.

1 Introduction

Near-field illumination need not decay monotonically with depth: attenuation and incidence angle change together, so one intensity can correspond to two depths. We seek the least coaxial-to-off-axis coefficient ratio that guarantees strict decrease at every positive depth over a prescribed layout and normal class.

A reference patch with calibrated normal and gain can be translated along the camera axis while its orientation remains fixed. With calibrated lighting, its distance z is decoded from one scalar intensity. Monotonicity gives a unique calibration; a coefficient margin bounds its sensitivity.

At a 45° normal-cone angle, the sharp worst-case isotropic ratios are $1/(4\sqrt{2}) \approx 0.17678$ for unrestricted layouts and $1/(25\sqrt{5}) \approx 0.017889$ for balanced common-radius layouts. Their ratio is $25\sqrt{10}/8 \approx 9.8821$ in calibrated source coefficients.

The balance bound holds across changes in the lit-source set. A weighted-centroid residual quantifies imperfect balance. At the critical coefficient,

the inverse can be non-Lipschitz despite uniqueness; a positive margin bounds depth error on a finite band.

A second observation changes the calibration requirements. For balanced common-radius arrays with nonnegative incidence, two coefficient allocations at unchanged depth, normal and gain eliminate the latter two inputs. With partial clamping, distinct depths can give the same data for every mixture of the coaxial source and one fixed off-axis pattern.

Contributions.

1. Theorems 1 and 2 give sharp isotropic thresholds for unrestricted and balanced common-radius layouts. Corollary 2 quantifies the effect of a weighted-centroid residual.
2. Theorem 3 bounds inverse sensitivity under a positive coefficient margin. Proposition 1 gives a non-Lipschitz inverse at the threshold. Fixed-budget allocations propagate bounded normal and gain errors.
3. Theorem 4 reduces the aligned angular threshold to a finite set of candidates. Corollary 3 gives its linear growth with normal-cone slope for every fixed positive emission exponent.
4. Proposition 2 recovers depth from two observations without normal or gain inputs for balanced arrays with nonnegative incidence. A clamped collision gives identical data at distinct depths for every mixture of one fixed off-axis pattern and the coaxial source.

2 Relation to near-field photometric methods

We use the near-field Lambertian model of LED photometric stereo [1], including point-source attenuation, local incidence, and clamping. Equations (2.1)–(2.2) of the 2017 author manuscript give the aligned angular source model; setting the emission exponent to zero gives the isotropic case. That work also fits a scalar shape scale by reprojection (Eq. (3.12)).

Spence and Chantler [2] optimize separate distant illuminations for photometric-stereo normal recovery. Prados and Faugeras [3] study a camera-centered source through a perspective shape-from-shading PDE and boundary state constraints.

Wu et al. [4] model simultaneous near-field sources by an additive Lambertian reflectance map [Eqs. (1), (9)–(10)] and jointly estimate depth and its image-plane derivatives by regularized variational fitting [Eqs. (12)–(17)]. Wetzler et al. [5] use separately acquired photometric equations with image ratios and characteristic reconstruction. The two-observation extension below uses such radiometric elimination to examine the calibration requirements of the scalar model.

Zhou and Tan [6] recover normals and lighting from a multi-image factorization [Eq. (1)]. Their Propositions 1–3 characterize and resolve the ring-light ambiguity; their prototype also accounts for distance-dependent cone angles (Sec. 5.1). Liu et al. [7] use circularly placed near point sources with sourcewise clamping [Eq. (3)], differential images, and mesh-depth fitting [Eqs. (7), (12)]. McGunnigle et al. [8] use coaxially augmented photometric stereo [9] for robust slope estimates and laser height profiles to recover discontinuous surfaces. Usami et al. [10] use a metric reference and calibrated two-source geometry for endoscopic shape recovery.

Symmetric source pairs cancel transverse normal terms before clamping [4]. We analyze the summed depth response after each source term has been clamped. The normal and gain are fixed, while positive off-axis strengths and camera-plane positions vary. The sharp threshold ranges over this layout class, including coincident and common-azimuth sources. Balanced common-radius arrays have a smaller, tilt-independent threshold.

3 Calibration model

The camera and the coaxial source are at the origin. The observed point is $X(z) = (0, 0, z)$, with $z > 0$. Its unit normal is

$$n = (b_1, b_2, -\nu), \qquad \nu > 0, \qquad \|b\|/\nu \le T < \infty. \tag{1}$$

Here $b = (b_1, b_2)$ is the transverse normal component, and $T = \tan\theta$ specifies a normal cone of half-angle $0 \le \theta < \pi/2$. There are finitely many off-axis sources, indexed by $i = 1, \dots, N$ with $N \ge 1$. Source i is at $(a_i, 0)$ in the camera plane, with $r_i = \|a_i\| > 0$ and coefficient $m_i > 0$. Write $m_0 \ge 0$ for the coaxial coefficient, $S = \sum_i m_i > 0$ for the off-axis total, and $h_i = b^\top a_i$. The isotropic

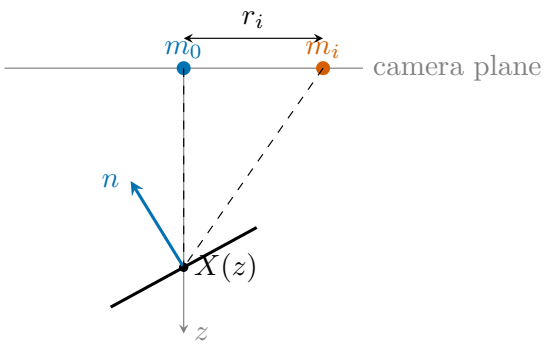


Fig. 1 An optical-axis calibration point under coaxial and off-axis illumination. The normal and source positions are fixed while the candidate depth varies.

gain-normalized response is

$$\Phi_n(z) = \frac{\nu m_0}{z^2} + \sum_i m_i \frac{[\nu z + h_i]_+}{(z^2 + r_i^2)^{3/2}}. \tag{2}$$

The positive part $[s]_+ = \max\{s, 0\}$ implements local Lambertian clamping. The measured intensity is $I(z) = \gamma \Phi_n(z)$, where the gain $\gamma > 0$ collects fixed surface and camera factors. The coefficients are calibrated source strengths. Sections 4–8 take n and γ as known and fixed; Section 9 considers them unknown but unchanged between exposures.

4 Sharp universal coefficient

Lemma 1 (Strictness across visibility transitions) *Let $f : (0, \infty) \to \mathbb{R}$ be locally absolutely continuous, with $f' \le 0$ almost everywhere. Suppose there is a finite set B such that f is real analytic and nonconstant on every component of $(0, \infty) \setminus B$. Then f is strictly decreasing.*

Proof Local absolute continuity first gives nonincrease. If $f(x) = f(y)$ for $x < y$, nonincrease makes f constant on $[x, y]$. Removing finitely many points leaves an open subinterval inside one analytic component. The identity theorem would make f constant on that component, a contradiction. □

For the isotropic model, the visibility breakpoints are the positive values $-h_i/\nu$. On a component with lit-index set L, the response agrees with the analytic function

$$F_L(z) = \frac{\nu m_0}{z^2} + \sum_{i \in L} m_i \frac{\nu z + h_i}{(z^2 + r_i^2)^{3/2}}.$$

This continuation is defined for all $z > 0$, without imposing the component's visibility conditions elsewhere. Its off-axis terms stay bounded as $z \downarrow 0$, whereas the coaxial term diverges when $m_0 > 0$. Thus no branch is constant. The isotropic thresholds below imply $m_0 > 0$; their derivative bounds and local absolute continuity then allow Lemma 1 to establish strict decrease, including equality and visibility transitions.

For $t > 0$, define $g_T(t) = t^3(1 - 2t^2 + 3Tt)/(1 + t^2)^{5/2}$.

Theorem 1 (Universal monotonicity threshold) *For $0 \le T \le 1$, let t_T be the unique positive root of*

$$1 + 4Tt - 4t^2 - Tt^3 = 0. \tag{3}$$

Set

$$c(T) = \begin{cases} \frac{1}{2} g_T(t_T), & 0 \le T \le 1, \\ \dfrac{T^3}{2(1 + T^2)^{3/2}}, & T \ge 1. \end{cases} \tag{4}$$

The smallest ratio that guarantees a strictly decreasing response for every allowed configuration is $c(T)$. Its sufficient condition is

$$m_0 \ge c(T) S \tag{5}$$

For every smaller ratio there is an allowed layout and normal whose response is not injective.

Proof On a lit interval, put $t = z/r_i$ and $u = h_i/(\nu r_i)$. Then $u \ge -T$, $t + u > 0$, and the normalized derivative of one source is

$$g(t, u) = \frac{t^3(1 - 2t^2 - 3ut)}{(1 + t^2)^{5/2}}. \tag{6}$$

It decreases in u. Its supremum at fixed t is $t^3/(1 + t^2)^{3/2}$ for $t \le T$, approached at $u = -t$, and $g_T(t)$ for $t \ge T$. The former function increases. Direct differentiation gives

$$g_T'(t) = \frac{3t^2(1 + 4Tt - 4t^2 - Tt^3)}{(1 + t^2)^{7/2}}. \tag{7}$$

For $T > 0$ the descending coefficient signs of the numerator polynomial have one variation, so there is one positive root. The $T = 0$ root is $1/2$. Since the polynomial at $t = T$ is $1 - T^4$, its root lies above T for $T < 1$ and below T for $T > 1$. This proves

$\sup g(t,u) = 2c(T)$, including the common value at $T = 1$.

Unlit source terms have zero derivative almost everywhere. Therefore

$$\frac{z^3}{\nu}\Phi_n'(z) \le -2m_0 + 2c(T)S. \tag{8}$$

Each clamped term is locally absolutely continuous on positive depth bands. The derivative bound and Lemma 1 give strict decrease, including equality and visibility transitions.

For necessity, place all off-axis strength at a common radius and azimuth, opposite the transverse normal, so that $u = -T$ for each source. For $T < 1$ the supremum occurs on a lit interval; for $T \ge 1$ approach $t = T$ from the lit side. A smaller ratio makes the derivative positive on an interval. The nonnegative response tends to zero at infinity, so this increasing interval is incompatible with injectivity of a continuous function on $(0, \infty)$. □

Corollary 1 (Endpoint ratios) *The normal-incidence and 45° ratios are*

$$c(0) = \frac{1}{25\sqrt{5}}, \qquad c(1) = \frac{1}{4\sqrt{2}}, \qquad \lim_{T\to\infty} c(T) = \frac{1}{2}. \tag{9}$$

All are ratios to the off-axis total S, not to total emitted strength $S + m_0$.

The flat-normal ratio cannot be reused for tilted normals. For example, take $T = 1$, $r = 2$, $z = 3$, $u = -1$, $S = 1$, and $m_0 = 1/50$. Although $1/50 > c(0)$, one has $g(3/2, -1) = 108/(169\sqrt{13}) > 1/25 = 2m_0$. The intensity derivative is positive on a strictly lit interval.

A check for a known operating band.

For the isotropic model, let $0 < z_{\min} < z_{\max}$ specify an operating band and $r_{\max} = \max_i r_i$. Put

$$t_{\mathrm{dec}}(T) = \frac{3T + \sqrt{9T^2 + 8}}{4}.$$

If $z_{\min} \ge r_{\max} t_{\mathrm{dec}}(T)$, then $m_0 = 0$ suffices for strict decrease on the band. Indeed, $t = z/r_i \ge t_{\mathrm{dec}}(T) > T$ places every source on a lit branch, and $1 - 2t^2 + 3Tt \le 0$, strictly for $z > z_{\min}$. Equation (6) makes every off-axis derivative negative in the band interior. At $T = 1$, $t_{\mathrm{dec}}(1) \approx 1.7808$. A specified operating band can therefore be checked before allocating a coaxial coefficient.

5 Isotropic balanced arrays remove the tilt penalty

Theorem 2 (Common-radius balance) *Suppose all off-axis lights have a common radius r and $\sum_i m_i a_i = 0$. Then $m_0 \ge c(0)S$ guarantees strict monotonicity for every calibrated fixed normal with $\nu > 0$, independently of its tilt. This ratio is sharp for the common-radius balanced array when the normal class contains the front-facing normal. A strict margin has the stability bound of Theorem 3, with $c(T)$ replaced by $c(0)$.*

Proof Balance implies $\sum_i m_i h_i = 0$. The derivative of the unclamped off-axis sum is $\nu S(r^2 - 2z^2)/(r^2 + z^2)^{5/2}$. An unlit term has $h_i \le -\nu z$, so its unclamped derivative numerator is at least $\nu(r^2 + z^2) > 0$. Clamping removes that positive derivative. Hence the physical off-axis derivative is no larger than the unclamped sum almost everywhere, and $z^3\Phi_{\mathrm{off}}'(z)/\nu \le S g_0(z/r) \le 2c(0)S$. The coaxial term gives the derivative bound, and Lemma 1 gives strictness also at equality. With a front-facing normal all sources are lit and the response is the flat one-radius response, proving sharpness over the normal class. □

Equal-strength antipodal pairs and uniform rings satisfy the condition. Separately balanced radial groups also inherit this sufficient bound.

Corollary 2 (Imperfect common-radius balance) *In the isotropic model, suppose all off-axis sources have radius $r > 0$ and*

$$\varepsilon = \frac{\|\sum_i m_i a_i\|}{Sr} \le \bar{\varepsilon} \le 1, \qquad \delta = T\bar{\varepsilon} \le 1.$$

Then $m_0 \ge c(\delta)S$ guarantees strict decrease for every fixed calibrated normal in the declared cone. This ratio is sharp over such layouts with arbitrary nonempty finite source sets and freely chosen positive strengths.

Proof Put $\bar{h} = S^{-1}\sum_i m_i h_i \ge -\nu r\delta$ and $t = z/r$. The clamping argument of Theorem 2 gives, almost everywhere,

$$\frac{z^3\Phi_n'(z)}{\nu S} \le -2\frac{m_0}{S} + g_\delta(t) \le -2\frac{m_0}{S} + 2c(\delta).$$

The last step uses $\delta \le 1$ and the maximization in Theorem 1. Since $c(\delta) > 0$, the ratio gives $m_0 > 0$; Lemma 1 also gives strictness at equality. For sharpness, take two sources of strength $S/2$ at $a_\pm = r(-\bar{\varepsilon}, \pm\sqrt{1 - \bar{\varepsilon}^2})$ and a unit normal with $b = (\nu T, 0)$.

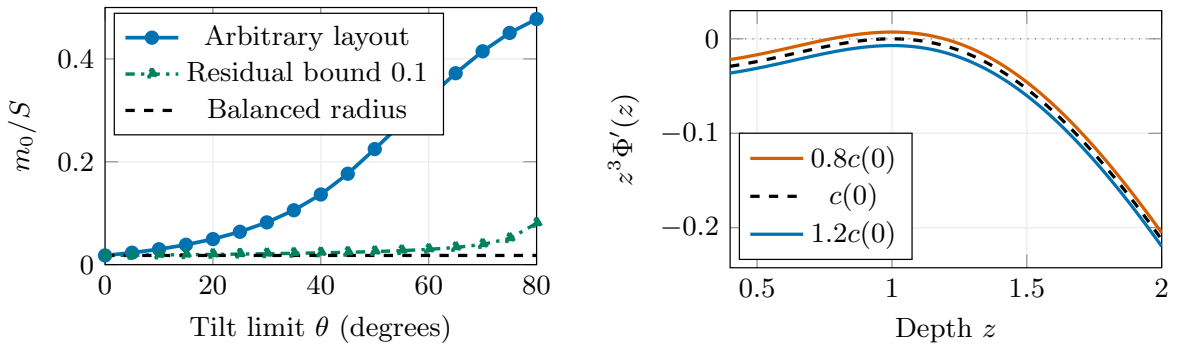


Fig. 2 Left: sharp isotropic coefficient ratios for unrestricted layouts, exact common-radius balance, and imperfect common-radius balance with $\bar{\varepsilon} = 0.1$. The residual curve uses $c(T\bar{\varepsilon})$ for normal-cone half-angle θ, with $T = \tan\theta$ and $T\bar{\varepsilon} \le 1$. Right: the flat-normal derivative for one unit off-axis source of radius two. Below the threshold an increasing interval appears; at equality the derivative touches zero; a strict margin shifts it below zero. The angle curve evaluates Eq. (4).

Both have $u = -\delta$. For $\delta < 1$, the maximizing depth satisfies $t_\delta > \delta$; for $\delta = 1$, approach $t = 1$ from the lit side. Every smaller nonnegative ratio therefore gives an increasing interval; the zero-tail argument of Theorem 1 completes sharpness. □

With $m_0 = c(\delta)S + \eta$, $\eta > 0$, Eq. (10) holds with the original ν. A bound uniform over the normal cone uses its smallest axial component, $\nu_{\min} = 1/\sqrt{1+T^2}$.

6 Stable inversion and the critical boundary

Theorem 3 (Strict-margin stability) *If $m_0 = c(T)S + \eta$ with $\eta > 0$, then on $[z_{\min}, z_{\max}] \subset (0, \infty)$,*

$$|z - z'| \le \frac{z_{\max}^3}{2\gamma\nu\eta}|I(z) - I(z')|. \tag{10}$$

For unit normals in the cone, a uniform constant replaces ν by $1/\sqrt{1+T^2}$.

Proof Equation (8) gives $-I'(z) \ge 2\gamma\nu\eta/z^3$ almost everywhere. Integrating between the two depths yields the result, including crossings of a local illumination boundary. □

Bounded model mismatch.

Let $B = [z_{\min}, z_{\max}]$ be a positive depth band and let I_{nom} satisfy Theorem 3. Suppose the actual response obeys

$$\sup_{s\in B} |\widetilde{I}(s) - I_{\text{nom}}(s)| \le \epsilon_{\text{model}}.$$

For a true depth $z \in B$, observe $y = \widetilde{I}(z) + e$ with $|e| \le \epsilon_{\text{meas}}$. Both error caps are nonnegative and use measured-intensity units. Set $J = [I_{\text{nom}}(z_{\max}), I_{\text{nom}}(z_{\min})]$ and $\widehat{z} = I_{\text{nom}}^{-1}(\Pi_J y)$, where Π_J clips to J. Then $\widehat{z} \in B$ and

$$|\widehat{z} - z| \le \frac{z_{\max}^3}{2\gamma\nu\eta}(\epsilon_{\text{meas}} + \epsilon_{\text{model}}). \tag{11}$$

Clipping cannot increase the distance to $I_{\text{nom}}(z) \in J$, so the bound follows from the triangle inequality and Eq. (10). It applies to each of the threshold models with its corresponding positive margin. The inverse is that of I_{nom}; the actual response $\widetilde{I}$ may be nonmonotone.

Proposition 1 (Injective but non-Lipschitz at equality) *Take a front-facing normal, unit gain, one off-axis source with strength one and radius two, and $m_0 = c(0)$. The response is strictly decreasing, but its inverse is not Lipschitz near the intensity at $z = 1$.*

Proof Direct differentiation gives $\Phi'(1) = \Phi''(1) = 0$ and

$$\Phi'''(1) = -\frac{96}{125\sqrt{5}}. \tag{12}$$

Consequently $\Phi(1+h) - \Phi(1) = -16h^3/(125\sqrt{5}) + O(h^4)$. The inverse depth error has cubic-root order in intensity error, precluding a finite local Lipschitz constant. This example proves that equality in the monotonicity threshold is not a universal stability guarantee. □

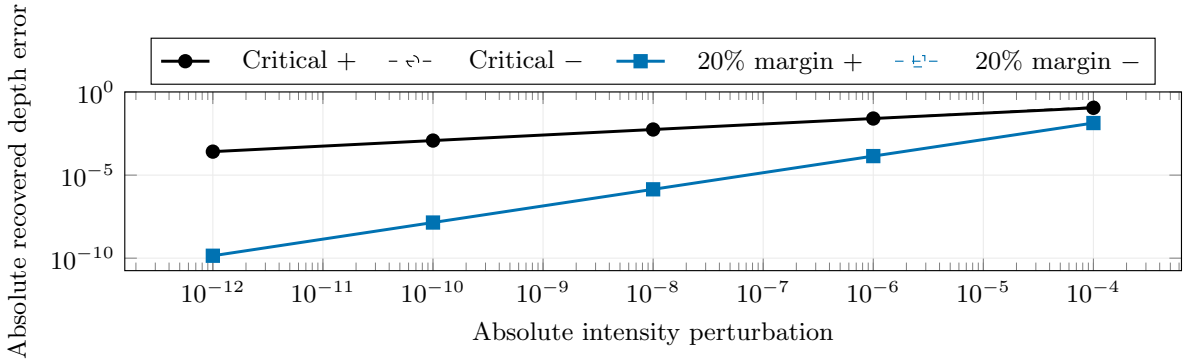


Fig. 3 Depth errors from high-precision inversion at $z = 1$, using the critical isotropic flat-normal configuration and a 20% coefficient margin. Each curve contains five perturbation magnitudes. Positive and negative perturbations give distinct errors, though their curves nearly overlap on these logarithmic axes, especially with the margin. The endpoint slopes show cubic-root scaling at the critical coefficient and linear scaling with the margin.

7 Aligned angular LED emission

The angular source model of Quéau et al. [1] also permits an exact design threshold. Suppose every source principal axis is parallel to the positive optical axis, and all sources have a common fixed emission exponent $\mu \geq 0$. An off-axis source contributes

$$m_i \frac{z^{\mu}[\nu z + h_i]_+}{(z^2 + r_i^2)^{(\mu+3)/2}}. \tag{13}$$

Setting $\mu = 0$ recovers the isotropic model. The coaxial contribution remains $\nu m_0/z^2$. The coefficients specify calibrated axial source strengths; total radiant power depends on the emission exponent.

On a visibility component, replace the off-axis summand in F_L by $m_i z^{\mu}(\nu z+h_i)/(z^2+r_i^2)^{(\mu+3)/2}$. For each fixed $\mu \geq 0$ these branches are analytic on $z > 0$ and their off-axis terms stay bounded at zero. Consequently, once the threshold below gives $m_0 > 0$ and nonincrease, the same nonconstancy argument and Lemma 1 give strict decrease across visibility transitions.

For a lit source, set $t = z/r_i$, $u = h_i/(\nu r_i)$, and define

$$G_{\mu}(t,u) = \frac{t^{\mu+2}\{t[(\mu+1) - 2t^2] + u[\mu - 3t^2]\}}{(1+t^2)^{(\mu+5)/2}}. \tag{14}$$

This is its derivative multiplied by $z^3/(\nu m_i)$. Write $s = \sqrt{\mu/3}$ and $B_{\mu}(t) = (t^2/(1+t^2))^{(\mu+3)/2}$. The three geometric branches below come from $u = T$, the illumination-boundary limit $u = -t$, and $u = -T$.

Theorem 4 (Angular-emission threshold) *In the aligned-axis model, the smallest universal ratio ensuring strict monotonicity is $m_0 \geq \kappa_{\mu}(T)S$, where*

$$2\kappa_{\mu}(T) = \max\Big\{0,\ \sup_{0<t\leq s} G_{\mu}(t,T),\ B_{\mu}(T) \text{ if } T \geq s,\ \sup_{t\geq \max(s,T),\ t>0} G_{\mu}(t,-T)\Big\}. \tag{15}$$

The first interval is omitted when $\mu = 0$. A closed endpoint on $t+u = 0$ denotes a limit from the lit side. Each fixed-u supremum is determined by its interval endpoints and the positive roots, lying in that interval, of

$$P_{\mu,u}(t) = (\mu+3)t[(\mu+1) - 4t^2] + u[\mu(\mu+2) - 6(\mu+2)t^2 + 3t^4]. \tag{16}$$

The endpoint limits at zero and infinity are 0 and -2, respectively. Thus only positive real roots of the polynomials of degree at most four and the branch endpoints are needed. A strict margin $m_0 = \kappa_{\mu}(T)S + \eta$ has the finite-band inverse bound of Theorem 3.

Proof The coefficient of u in G_{μ} has the sign of $\mu - 3t^2$. For $0 < t \leq s$, its maximum over $|u| \leq T$ and $t+u > 0$ occurs at $u = T$. For $s < t \leq T$, the supremum is approached at $u = -t$, giving $B_{\mu}(t)$, which increases in t. For $t > \max(s,T)$, the maximizing choice is $u = -T$.

This proves Eq. (15). Direct differentiation yields

$$\partial_t G_\mu(t,u) = \frac{t^{\mu+1} P_{\mu,u}(t)}{(1+t^2)^{(\mu+7)/2}}. \tag{17}$$

The polynomial is nonzero because its cubic coefficient is $-4(\mu+3)$. Its positive real roots and the endpoint limits determine the candidate values.

The choice $u = 0$ has a positive maximum, so $\kappa_\mu(T) > 0$. Every clamped source satisfies $z^3\Phi_i'(z)/\nu \le 2m_i\kappa_\mu(T)$ almost everywhere. Summing with the coaxial derivative proves nonincrease at the stated ratio. Lemma 1 gives strict decrease also at equality and across visibility transitions.

For any smaller ratio, choose a strictly lit (t,u) whose derivative exceeds $2m_0/S$. A unit normal with transverse-to-axial ratio T and a suitable source azimuth realizes any $u \in [-T,T]$; use $u = 0$ when $T = 0$. Concentrating all off-axis strength at that radius and azimuth makes the response increase on an interval. Its nonnegative zero tail then precludes injectivity. A boundary supremum is approached by strictly lit configurations. The strict-margin bound follows by the same integration as Theorem 3. □

At $T = 0$, $\kappa_\mu(0) = ((\mu+1)/(\mu+5))^{(\mu+5)/2}$; the flat case maximizes at $t = \sqrt{\mu+1}/2$. In the isotropic case, $\kappa_0(T) = c(T) \to 1/2$.

Corollary 3 (Asymptotic angular slope) *For every fixed $\mu > 0$, let $s = \sqrt{\mu/3}$ and*

$$C_\mu(t) = \frac{t^{\mu+2}(\mu - 3t^2)}{(1+t^2)^{(\mu+5)/2}}.$$

Then

$$\lim_{T\to\infty} \frac{\kappa_\mu(T)}{T} = A_\mu := \frac{1}{2} \max_{0<t<s} C_\mu(t) > 0.$$

The maximum is attained uniquely at $t = \sqrt{x_\mu}$, where

$$x_\mu = \mu + 2 - \sqrt{\tfrac{2}{3}(\mu+2)(\mu+3)} \in (0, \mu/3).$$

Proof Write $G_\mu(t,u) = a_\mu(t) + uC_\mu(t)$, where $a_\mu(t) = G_\mu(t,0)$. Both functions extend continuously to $[0,s]$; C_μ is zero at the endpoints and positive inside. Hence $(a_\mu + TC_\mu)/T$ converges uniformly to C_μ there. For $t \ge s$, the coefficient $C_\mu(t)$ is nonpositive. Every lit choice $u > -t$ therefore satisfies

$$G_\mu(t,u) \le G_\mu(t,-t) = B_\mu(t) \le 1.$$

With $F_T = \sup_{0<t\le s} G_\mu(t,T)$, the envelope in Eq. (15) consequently gives $F_T \le 2\kappa_\mu(T) \le \max\{F_T, 1\}$. Dividing by T proves the limit. Finally, $C_\mu'(t)$ has the sign of $Q_\mu(x) = 3x^2 - 6(\mu+2)x + \mu(\mu+2)$ with $x = t^2$. On $[0, \mu/3]$, this polynomial is strictly decreasing, positive at zero, and negative at $\mu/3$. Its unique root is x_μ, proving the stated maximizer. □

The isotropic balance theorem is geometry-specific. With angular emission, deleting an unlit term need not remove a positive unclamped derivative. For $\mu = 1$, an equal antipodal pair at radius one, $S = 1$, $\nu = 1/\sqrt{65}$, $b = (8/\sqrt{65}, 0)$, $z = 1/2$, and $m_0 = 1/27$ gives $z^3\Phi'(z)/\nu = 47/3375 > 0$. Thus the flat angular coefficient cannot replace $c(0)$ in a tilt-independent balanced-array guarantee.

7.1 Finite-candidate evaluation of the angular coefficient

For $s = \sqrt{\mu/3}$, retain positive real roots of $P_{\mu,T}$ in $0 < t \le s$ when $\mu > 0$, and roots of $P_{\mu,-T}$ with $t \ge \max(s,T)$ and $t > T$. Include the finite branch endpoints, the limits 0 and -2 at zero and infinity, and $B_\mu(T)$ when $T \ge s$. Half the largest candidate gives $\kappa_\mu(T)$ by Eq. (15). Shadow-boundary values are lit-side limits. Online Resource 1 gives the full enumeration and the implementation for $\mu = 1$.

8 Numerical validation

We compare the reduced formulas with a separate three-dimensional implementation. For each source, it forms the direction vector from the calibration point, normalizes it, clamps the normal–direction inner product, and applies inverse-square attenuation. All calculations use the radiometric models above. The isotropic cases span four tilt parameters, three layouts, and five depths. Another 40 cases check the tilt-independent bound for an equal-strength pair and a four-source balanced ring. Twelve extremal-layout cases compare coefficients below, at, and above the threshold; those with $T \ge 1$ approach the illumination boundary from the lit side.

Computed coefficients, depth errors and endpoint slopes are rounded to five significant figures; calculations use the full-precision values in Online Resource 1.

For the inversion check, we perturb the intensity at the flat-normal critical point and recover depth by 80-digit bisection on $[0.5, 1.5]$. We use both perturbation signs at five magnitudes, comparing the critical coefficient with a 20% margin.

Imperfect balance.

At 45° and $\bar{\varepsilon} = 0.1$, the required class-wide ratio is 0.024061, or $1.3451\,c(0)$. Across 21 cases,

Property	Calculation	Result
Geometry, attenuation, incidence	Independent 3D versus reduced irradiance, 60 cases	1.11×10^{-16} maximum error
Derivative and clamp sign	Finite difference versus analytic derivative, 60 cases including one illumination boundary	6.26×10^{-10} maximum error
Balance cancellation	Equal-strength pair and balanced ring, 40 cases	40/40 bounds satisfied
Sharpness	Extremal layouts at coefficient factors below, at, and above threshold, 12 cases	12/12 signs as predicted
Finite-band stability	Strict-margin inverse bound, 10 cases	10/10 satisfied
Monotonicity versus stability	Critical inversion, both noise signs	Slopes 0.32973–0.33715
Strict-margin scaling	Same inversion with 20% margin, both signs	Slopes 0.99880–1.0011
Coordinate dependence	Angular base, joint rotation, and normal-only rotation, 36 conditions	36/36 comparisons satisfied

Table 1 Numerical checks of the radiometric formulas, thresholds, and inverse response. A second-order right difference is used at the single illumination-boundary case; the other 59 use a central difference. Endpoint log slopes use the two endpoint noise magnitudes.

maximum irradiance and normalized-derivative discrepancies are 3×10^{-80} and 4.51×10^{-25}; the CSV inputs and reproduction scripts are supplied in Online Resource 1.

Aligned angular emission.

The angular calculations use $\mu = 1$ and $T \in \{0, 1, 8, 64\}$. The $T = 64$ case covers the positive-u branch with a witness whose envelope exceeds the shadow-boundary value. For each chosen witness, coefficient factors $0.8, 1, 1.2$ are compared under the base geometry, joint 90° rotation, and normal-only 90° rotation. All 36 conditions satisfy the comparisons. The maximum absolute irradiance discrepancy is 2×10^{-80}, and the maximum finite-difference discrepancy in the normalized derivative is 7.50×10^{-16}.

From an error budget to a calibrated coefficient.

For the isotropic model, put $\nu_{\min} = 1/\sqrt{1+T^2}$. Given an intensity-error cap $\epsilon_I > 0$ and target depth error $\delta_z > 0$, choose

$$\eta = \frac{z_{\max}^3 \epsilon_I}{2\gamma \nu_{\min} \delta_z}, \qquad m_0 = c(T)S + \eta.$$

For combined measurement and model error, use $\epsilon_I = \epsilon_{\text{meas}} + \epsilon_{\text{model}}$. The margin η has the same units as m_0 and S. With the clipped inverse, this choice bounds depth error by δ_z throughout the band and normal cone.

The worked example uses $T = 1$, $S = \gamma = 1$, $[z_{\min}, z_{\max}] = [0.5, 3]$, $\epsilon_I = 10^{-4}$, and $\delta_z = 0.02$, all in the declared model units. It uses one off-axis source at $(2, 0, 0)$ with coefficient one and the known normal $(-1, 0, -1)/\sqrt{2}$. Thus $\nu_{\min} = 1/\sqrt{2}$ and $c(1) = 1/(4\sqrt{2})$; the resulting coefficients are $\eta \approx 0.095459$ and $m_0 \approx 0.27224$. Since $\gamma = 1$, measured and gain-normalized intensity errors coincide. The inverse bound has constant 200, so $200\epsilon_I = 0.02$. We generate intensities with the coordinate-form evaluator, clip noisy values to the endpoint-intensity interval, and invert the reduced formula by 80-digit bisection.

Table 3 compares the two coefficient choices at five depths and both noise signs, giving twenty observations. The threshold-only choice retains monotonicity but has no positive strict margin; its maximum observed error is 0.27288 and six of ten rows meet the target. The margin design meets the target in all ten rows, with maximum error 0.010387.

8.1 Fixed budgets and calibration uncertainty

Let η_{req} be the margin required by the preceding error-budget formula. For a total coefficient budget $M = m_0 + S$, substituting $S = M - m_0$ gives

$$\frac{m_0}{M} = \frac{c + \lambda}{1 + c}, \qquad 0 < \lambda = \frac{\eta_{\text{req}}}{M} < 1. \tag{18}$$

T	$\kappa_1(T)$ (numerical)	Maximizing candidate	Positive roots $(+T/-T)$
0	0.037037	$u=-T$ interior	1/1
1	0.13517	$u=-T$ interior	2/2
8	0.48473	Shadow-boundary limit	2/2
64	0.70158	$u=T$ interior	2/2

Table 2 Angular results for $\mu=1$. Root counts concern the two fixed-u polynomials before branch filtering. The shadow-boundary value is a supremum; the forward checks use lit configurations approaching the boundary.

Choice	m_0	$E_{\max}$	Target met
Threshold	0.17678	0.27288	6/10
Margin design	0.27224	0.010387	10/10

Table 3 Coefficient selection for the isotropic model. Each choice has ten observations; $E_{\max}$ is their maximum absolute depth error. Coefficients and errors use model units.

Keep source positions and relative off-axis weights fixed as the budget is allocated. Use $c=c(T)$, $c(T\bar{\varepsilon})$, or $c(0)$ for the unrestricted, imperfect-balance, or balanced class, respectively.

To account for calibration error, treat n and γ as nominal inputs. Suppose the actual unit normal n' is fixed and camera-facing, $\|n'-n\|_2\le\rho_n$, and the actual gain is $\gamma'=\gamma(1+\xi)$ with $|\xi|\le\rho_\gamma<1$. Keep source positions and coefficients unchanged. With $\beta=\rho_n+\rho_\gamma$ and common off-axis radius r, local clamping gives

$$|I_{\mathrm{true}}(z)-I_{\mathrm{nom}}(z)|\le\gamma\beta A(z),\qquad A(z)=\frac{m_0}{z^2}+\frac{S}{z^2+r^2}. \tag{19}$$

For a target depth tolerance $0<\delta_z<z_{\max}-z_{\min}$, put $z_*=z_{\max}-\delta_z$ and $H=z_*^{-2}-z_{\max}^{-2}$. The nominal clipped inverse has error at most δ_z throughout the band if

$$\frac{\epsilon_{\mathrm{meas}}}{\gamma}+\beta A(z_*)\le\nu\eta H,\qquad \eta=m_0-cS>0. \tag{20}$$

Online Resource 1 derives the certificate and the corresponding fixed-budget allocation rule.

The study uses nominal normal $n=(-1,0,-1)/\sqrt{2}$ and fixes $M=\gamma=1$, $T=1$, $r=2$, $[z_{\min},z_{\max}]=[0.5,3]$, $\epsilon_{\mathrm{meas}}=10^{-4}$ and $\delta_z=0.02$. With exact nominal calibration, Eq. (18) gives coaxial fractions 0.23134, 0.11671 and 0.11136 for the unrestricted, residual-0.1 and balanced classes, respectively.

Across the three layout classes, we evaluate 36 settings from normal-distance caps $\{0,0.0002,0.002,0.02\}$ and relative-gain caps $\{0,0.001,0.01\}$. Equation (20) certifies allocations below the budget for eighteen settings. Each scheme has 1,152 controls on these same settings: maximum depth errors are 0.032837 for nominal allocations and 0.014970 for certified allocations, with four and zero target exceedances. The certificates use exact rational arithmetic; separate direct-vector and reduced responses and 80-digit inversion provide the numerical controls. Online Resource 1 retains the eighteen uncertified settings and all 2,304 nominal-allocation controls.

Table 4 gives a representative cap pair. The nominal residual and balanced allocations also meet the target in these listed controls, illustrating conservatism of the uncertainty-uniform rule. At the chosen error target, the rule reaches its budget limit at $\beta\approx0.0085086$; the gray region in Fig. 4 shows where it certifies no allocation below the budget.

9 Observation count and calibration inputs

With a free positive gain, a single intensity no longer determines depth. We therefore consider two exposures of the same point at unchanged depth, normal and gain. Table 5 summarizes the calibration requirements.

Let the off-axis sources have common radius r and fixed positive relative weights w_i, with $\sum_i w_i=1$ and $\sum_i w_i a_i=0$. Exposure j has coaxial coefficient m_{0j} and off-axis coefficients $S_j w_i$, where $S_j=M-m_{0j}$. Each exposure uses coefficient budget M, giving $2M$ across the pair. Source geometry and relative illumination coefficients remain calibrated.

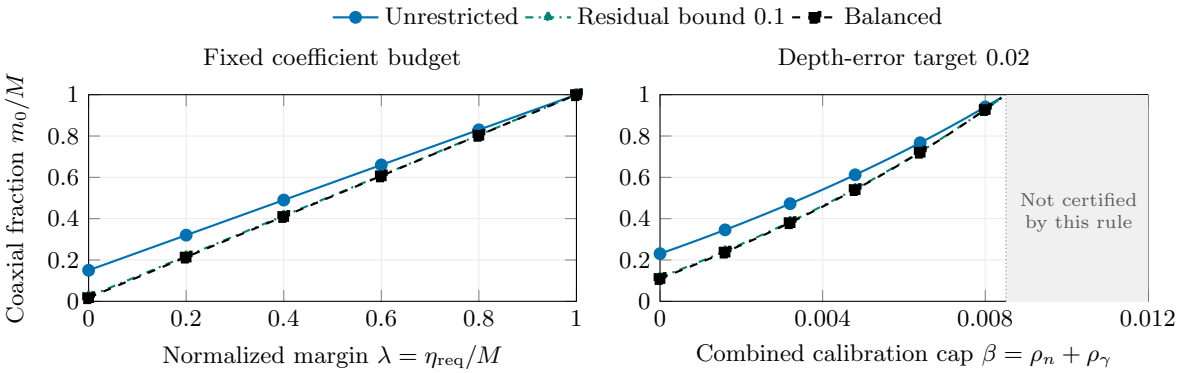


Fig. 4 Fixed-budget coaxial allocation m_0/M, where $M = m_0 + S$. Left: Eq. (18) at $T = 1$, with $\lambda = \eta_{\mathrm{req}}/M$; endpoints $\lambda = 0, 1$ are limiting cases. Right: Eq. (20) for depth tolerance 0.02 and $\beta = \rho_n + \rho_\gamma$, where ρ_n bounds unit-normal distance and ρ_γ relative gain error. Gray denotes caps not certified by this rule. Other parameters are given in Sec. 8.1.

Class	Nominal m_0/M	Certified m_0/M	Nominal $E_{\max}$	Certified $E_{\max}$
Unrestricted	0.23134	0.45600	0.032837	0.013866
Residual bound 0.1	0.11671	0.36618	0.013391	0.0097071
Balanced	0.11136	0.36192	0.012144	0.0092417

Table 4 Fixed-budget calibration controls at $\rho_n = 0.002$ and $\rho_\gamma = 0.001$. Nominal allocations use Eq. (18); certified allocations satisfy Eq. (20). $E_{\max}$ is the maximum absolute depth error in 64 boundary-perturbation controls per class and allocation. Full inputs, exact certificates and all rows are in Online Resource 1.

Proposition 2 (Two observations with unknown normal and gain) *Suppose* $0 < m_{01} < m_{02} < M$, *the common aligned emission exponent* $\mu \geq 0$ *is known, and every incidence factor* $\nu z + h_i$ *is nonnegative. Put* $k = \mu + 3$ *and* $D = m_{02}S_1 - m_{01}S_2 = M(m_{02} - m_{01}) > 0$. *The two intensities* I_1, I_2 *determine*

$$U = \frac{S_1 I_2 - S_2 I_1}{D}, \qquad V = \frac{m_{02} I_1 - m_{01} I_2}{D}, \tag{21}$$

and recover depth without the normal or gain as inputs:

$$q = V/U \in (0, 1), \qquad z = \mathcal{Z}(q) = \frac{r q^{1/k}}{\sqrt{1 - q^{2/k}}}. \tag{22}$$

For an otherwise unknown positive gain, two observations are minimal for exact depth identification on a nontrivial band of positive responses.

Proof Balance and nonnegative incidence give

$$I_j = U\{m_{0j} + S_j q(z)\}, \qquad U = \gamma\nu/z^2,$$

where $q(z) = (z/\sqrt{z^2 + r^2})^k$. The nonzero determinant gives Eq. (21), with $V = Uq(z)$, and inversion gives Eq. (22). The unknown normal and gain enter only through U. For one positive intensity and any candidate depth with positive forward response, a positive gain reproduces the measurement, even if the normal is fixed and known. Hence one intensity does not identify depth when its gain is free. □

The normal-cone bound ensures nonnegative incidence throughout a band with $z_{\min} \geq rT$. This supplies a range condition for using Eq. (22) without knowing the individual normal.

A collision when local clamping is allowed. Take $\mu = 0$, $r = M = 1$ and an equal pair at $(\pm 1, 0)$. Set $q_0 = 1/(5\sqrt{5})$ and $\sigma = 29\sqrt{145}/250 - 2/5$. The configurations

$$(z_1, n_1, \gamma_1) = (1/2, (0, 0, -1), 1), \qquad z_2 = 2/5, \quad n_2 = \frac{(\sigma, 0, -1)}{\sqrt{1 + \sigma^2}}, \qquad \gamma_2 = \tfrac{16}{25}\sqrt{1 + \sigma^2}$$

give the same unmixed values $U = 4$ and $V = 4q_0$. Here $2/5 < \sigma < 1$, so both normals lie in the $T = 1$ cone and one source is clamped in the

Observations	Calibration inputs	Illumination condition	Depth conclusion
One intensity	Normal and gain calibrated	General declared layout; local clamping allowed	Sharp coefficient threshold gives uniqueness; a strict margin bounds sensitivity
Two allocations	Normal and gain unknown but unchanged	Balanced common radius; all incidence factors nonnegative	Closed-form depth inverse; one positive intensity is insufficient
Any number of mixtures	Normal and gain unknown	Fixed balanced off-axis pattern; partial clamping allowed	Distinct-depth configurations can produce every mixture intensity identically

Table 5 Observation budget and calibration requirements. The last row is witnessed by the isotropic $T = 1$ construction in the text; it concerns mixtures of one fixed off-axis pattern and the coaxial source.

ϵ_{meas}	Controls	Maximum error	Maximum bound	Target met	Target certified
10^{-6}	432	0.0027600	0.0029818	432/432	432/432
10^{-4}	432	0.24545	0.30199	384/432	216/432

Table 6 Two-observation controls for the stated band and parameter grid. The target is 0.02. Certification uses Eq. (23); all observed errors lie below their individual bounds. Normal and gain are used only by the forward evaluator.

second configuration. Every mixture of the coaxial source and this fixed off-axis pattern therefore gives the same intensity $4(m_0 + Sq_0)$ at the two depths. For $m_{01} = 1/4$ and $m_{02} = 3/4$, the pair is $(1 + 3q_0, 3 + q_0)$. Both allocations exceed the unrestricted $T = 1$ monotonicity threshold. The collision persists for any number of such mixture exposures with the same relative off-axis weights. Any estimator based on these mixtures has worst-case absolute depth error at least 0.05 over this pair. Online Resource 1 gives the exact construction.

An observable error bound.

Let the true depth satisfy $z \in [z_{\min}, z_{\max}] \subset (0, \infty)$. Under the nonnegative-incidence assumptions of Proposition 2, take $m_{01} = M/4$, $m_{02} = 3M/4$ and observe $y_j = I_j + e_j$ with $|e_j| \le \epsilon_{\mathrm{meas}}$. Apply Eq. (21) to y_1, y_2 to obtain $\widehat{U}, \widehat{V}$, clip $\widehat{V}/\widehat{U}$ to $[q(z_{\min}), q(z_{\max})]$, and apply $\mathcal{Z}$. If $\widehat{U} > 0$, the resulting depth satisfies

$$|\widehat{z} - z| \le \frac{4L_q \epsilon_{\mathrm{meas}}}{M\widehat{U}}, \tag{23}$$

where L_q is the maximum of $\mathcal{Z}'$ on the clipping interval. This bound uses the observations and known band and illumination inputs. Online Resource 1 gives its endpoint formula and proof.

Numerical controls.

The numerical study uses $r = M = 1$, $[z_{\min}, z_{\max}] = [1, 3]$, $\mu \in \{0, 1\}$, an equal antipodal pair or a four-source ring, three normals in the $T = 1$ cone, gains $0.2, 1, 5$, and depths $1, 2, 3$. The decoder receives neither normal, gain nor true depth. All 108 noiseless controls recover depth with maximum error 1.1×10^{-78} at 80-digit precision. A separate ratio formula and 100-digit power evaluation agree within 1.53×10^{-78}.

For each noise cap, 432 controls cover all sign pairs. All 864 observed errors satisfy Eq. (23). Table 6 reports errors and certificates against target 0.02.

10 Discussion

The unrestricted threshold applies to unbalanced arrays and arrays with several radii. A fixed layout can need less coaxial strength. Common-radius balance gives a sharp reduction, and the weighted-centroid residual quantifies its loss under imperfect balance. The angular counterexample shows why the isotropic reduction depends on the emission model.

The threshold ensures uniqueness; the additional margin controls error amplification over the operating band. A particular array can have a steeper response than the class-wide derivative bound, permitting a smaller sufficient allocation. In the fixed-budget study, balance lowers

the nominal allocation, and calibration uncertainty reduces the allocation difference between classes. The two-observation result identifies a regime where normal and gain calibration can be removed. With partial clamping, different normals and gains can produce identical observations at different depths even above the monotonicity threshold. Off-axis observation points and nonlocal visibility require a corresponding forward model.

11 Conclusion

Sharp coefficient thresholds specify when illumination design makes scalar depth calibration unique. Common-radius balance reduces the isotropic requirement after local clamping, and the centroid residual quantifies imperfect balance. A positive margin controls error amplification; equality can leave a non-Lipschitz inverse. Angular emission changes the threshold and its grazing-tilt growth. Two observations eliminate normal and gain under nonnegative incidence, whereas a clamped collision shows why conditional monotonicity does not imply depth identification with unknown calibration inputs.

Data and code availability. Online Resource 1 contains the code, inputs, reference results, figure and animation sources, and reproduction instructions. The private companion repository is at: https://github.com/usamihiroyasu/coaxial-depth-calibration.

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References

[1] Y. Quéau, B. Durix, T. Wu, D. Cremers, F. Lauze, and J.-D. Durou, "LED-based photometric stereo: Modeling, calibration and numerical solution," *Journal of Mathematical Imaging and Vision*, vol. 60, no. 3, pp. 313–340, 2018. https://doi.org/10.1007/s10851-017-0761-1. Equation references use the author manuscript arXiv:1707.01018v2, 2017.

[2] A. D. Spence and M. J. Chantler, "Optimal illumination for three-image photometric stereo using sensitivity analysis," *IEE Proceedings: Vision, Image and Signal Processing*, vol. 153, no. 2, pp. 149–159, Apr. 2006. https://doi.org/10.1049/ip-vis:20050229.

[3] E. Prados and O. Faugeras, "Shape from shading: A well-posed problem?," in *Proc. IEEE CVPR*, vol. 2, pp. 870–877, 2005. https://doi.org/10.1109/CVPR.2005.319.

[4] C. Wu, S. G. Narasimhan, and B. Jaramaz, "A multi-image shape-from-shading framework for near-lighting perspective endoscopes," *International Journal of Computer Vision*, vol. 86, no. 2–3, pp. 211–228, 2010 (published online 12 February 2009). https://doi.org/10.1007/s11263-009-0207-3.

[5] A. Wetzler, R. Kimmel, A. M. Bruckstein, and R. Mecca, "Close-range photometric stereo with point light sources," in *Proc. International Conference on 3D Vision*, pp. 115–122, 2014. https://doi.org/10.1109/3DV.2014.68.

[6] Z. Zhou and P. Tan, "Ring-light photometric stereo," in *Computer Vision—ECCV 2010*, ser. Lecture Notes in Computer Science, vol. 6312. Springer, 2010, pp. 265–279. 10.1007/978-3-642-15552-9_20.

[7] C. Liu, S. G. Narasimhan, and A. W. Dubrawski, "Near-light photometric stereo using circularly placed point light sources," in *Proc. IEEE ICCP*, pp. 1–10, 2018. https://doi.org/10.1109/ICCPHOT.2018.8368465.

[8] G. McGunnigle, J. Dong, L. Su, and Y. Fang, "Recovering discontinuous surfaces with photometric stereo and laser sectioning," *Optical Engineering*, vol. 50, no. 5, article 053605, 2011. https://doi.org/10.1117/1.3575651.

[9] G. McGunnigle and J. Dong, "Augmenting photometric stereo with coaxial illumination," *IET Computer Vision*, vol. 5, no. 1, pp. 33–49, 2011. https://doi.org/10.1049/iet-cvi.2009.0038.

[10] H. Usami et al., "Polyp shape recovery from single endoscope image using medical suture," *The Open Bioinformatics Journal*, vol. 12, pp. 1–17, 2019. https://doi.org/10.2174/1875036201912010001.